

Assignments Continuous Integration Using Jenkins

Submitted to - Vikul Sir

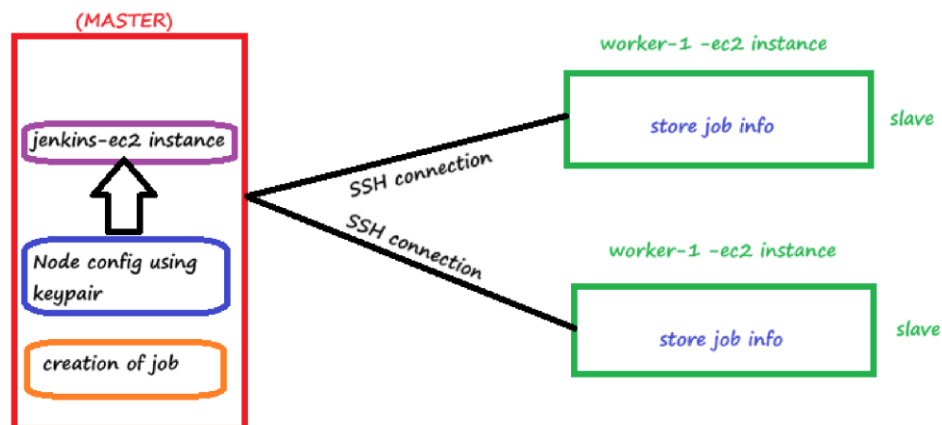
Date of Submission – 29/MAR/2026

Submitted by – NALATHATI RAHUL

L1 - Create Jenkins Master-Slave Configurations and Create a workspace in Jenkins Slave Node using Jenkins Free-style project

Step: 1 Architecture

Master and slave architecture in Jenkins



Step: 2 launch 3 instances

The screenshot shows the AWS Management Console for the 'us-east-1' region. The 'Instances' page displays three EC2 instances: 'master', 'worker-1', and 'worker-2'. All three instances are in the 'Running' state. The table below summarizes the instance details:

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IPv4 ...
master	i-0edbd9854bea38e11	Running	t3.micro	3/3 checks passed	View alarms +	us-east-1a	ec2-35-175-124-234.co...	35.175.124.23
worker-1	i-0fe5a066f6864600f	Running	t3.micro	3/3 checks passed	View alarms +	us-east-1a	ec2-3-236-161-252.co...	3.236.161.252
worker-2	i-008d0ab83a5135b4b	Running	t3.micro	3/3 checks passed	View alarms +	us-east-1a	ec2-98-80-172-99.com...	98.80.172.99

Below the table, the 'Monitoring' section shows various metrics for the selected instances, including CPU utilization, Network in (bytes), Network out (bytes), and Network packets in (count). The bottom right corner of the screenshot shows the system clock as 22:21 on 28/03/2026.

Step: 3 open MobXterm and connect to the servers

The screenshot shows the MobXterm application interface. The top bar displays '2 master', '1 worker-1', and '1 worker-2' sessions. The main terminal window shows the output of an SSH session to the 'master' instance. The output includes the SSH session details and the Amazon Linux 2023 logo and URL.

```
SSH session to ec2-user@35.175.124.234
• Direct SSH : ✓
• SSH compression : ✓
• SSH-browser : ✓
• X11-forwarding : ✗ (disabled or not supported by server)
• For more info, ctrl+click on help or visit our website.

#####
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023

[ec2-user@ip-172-31-0-152 ~]$
```

The bottom right corner of the screenshot shows the system clock as 22:27 on 28/03/2026.

Step: 4 install Prerequisite

On master server

→java installation

→git installation

→Jenkins installation

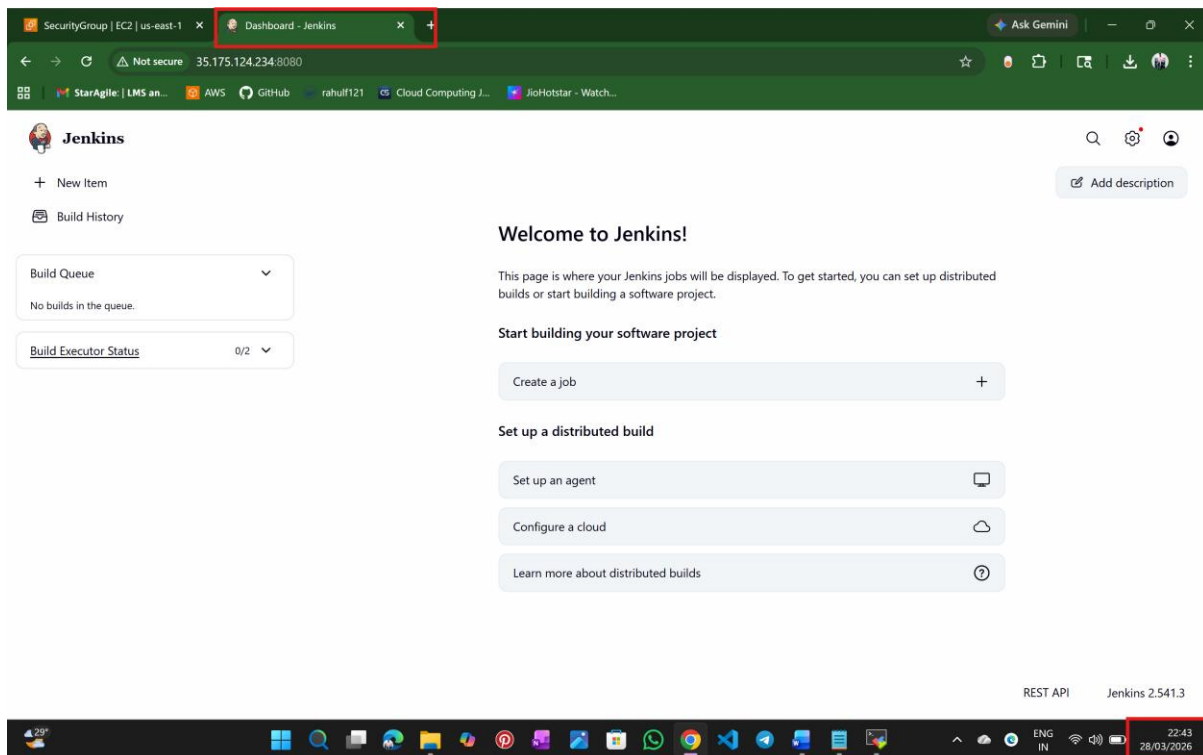
On workers 1 & 2

→→java installation

→git installation

→maven installation

Step: 5 logins to Jenkins



Step:6 configure worker nodes in Jenkins manager

Name?

Worker-1

Worker-2

Description?

Give Plain text

Number of executors?

Min (3) & max (5)

Remote root directory?

give directory name where to store build files

Labels?

Name the label (important

)

Usage?

Launch agent by connecting it to the controllerLaunch agents via SSH

Host?

Pvt ip address of the server

Credentials?

ec2-user (node-creds)

Add

Host Key Verification Strategy?

Non verifying Verification Strategy

Availability?

Keep this agent online as much as possible

Instances | EC2 | us-east-1

Nodes - Manage Jenkins - Jeni

Ask Gemini

35.175.124.234:8080/computer/

StarAgile | LMS an... AWS GitHub rahulf121 Cloud Computing J... JioHotstar - Watch...

Jenkins / Nodes

Nodes

+ New Node

Configure Monitors

Add, remove, control and monitor the various nodes that Jenkins runs jobs on.

S	Name	Architecture	Clock Difference	Free Disk Space	Free Swap Space	Free Temp Space	Response Time
	Built-In Node	Linux (amd64)	In sync	27.49 GiB	0 B	453.70 MiB	0ms
	worker-1	Linux (amd64)	In sync	27.83 GiB	0 B	458.24 MiB	16ms
	worker-2	Linux (amd64)	In sync	27.83 GiB	0 B	458.25 MiB	40ms
Data obtained		2.5 sec	2.5 sec	2.4 sec	2.4 sec	2.4 sec	2.4 sec

Icon: S M L

Legend

REST API Jenkins 2.541.3

22:56 28/03/2026

Instances | EC2 | us-east-1

master-slave Config - Jenkins

Ask Gemini

35.175.124.234:8080/job/master-slave/configure

StarAgile | LMS an... AWS GitHub rahulf121 Cloud Computing J... JioHotstar - Watch...

Jenkins / master-slave / Configure

Configure

General

Source Code Management

Triggers

Environment

Build Steps

Post-build Actions

Execute concurrent builds if necessary

☒ Restrict where this project can be run

Label Expression

node1

Label node1 matches 1 node. Permissions or other restrictions provided by plugins may further reduce that list.

Advanced

Source Code Management

Connect and manage your code repository to automatically pull the latest code for your builds.

☐ None

☒ Git

Repositories

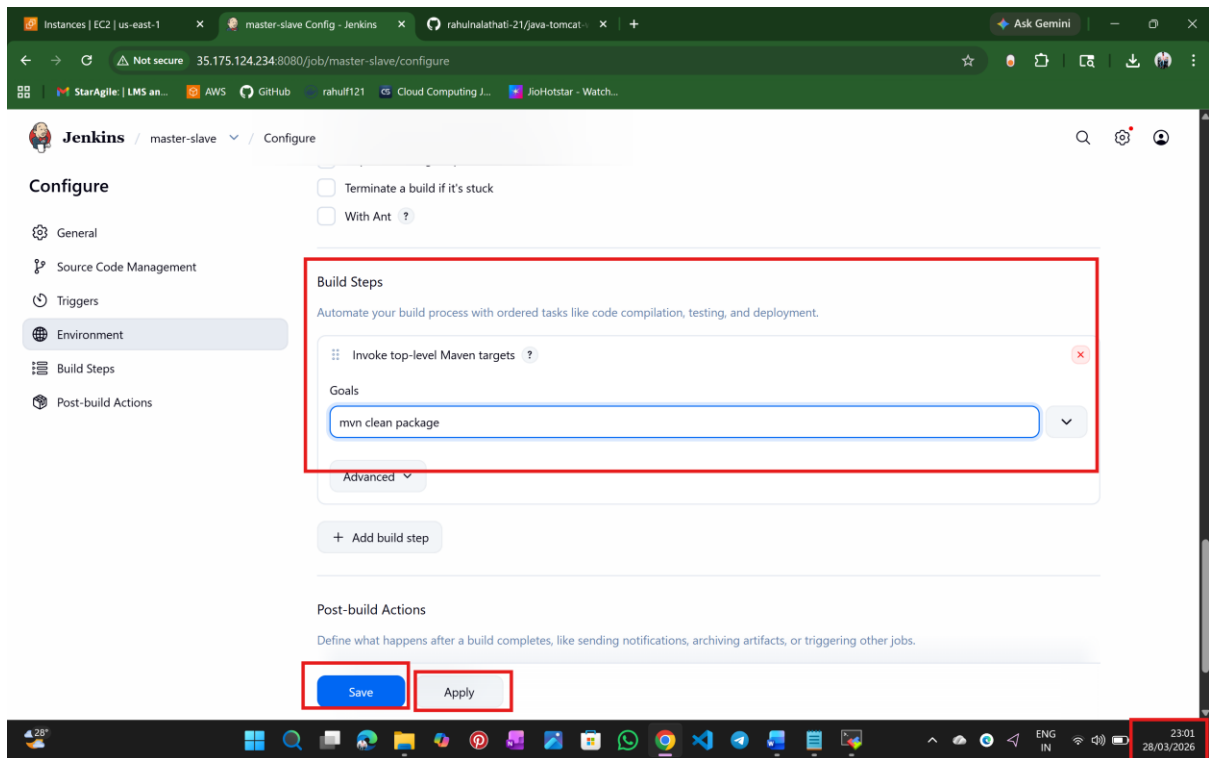
Repository URL

https://github.com/rahulnalathati-21/java-tomcat-webapp.git

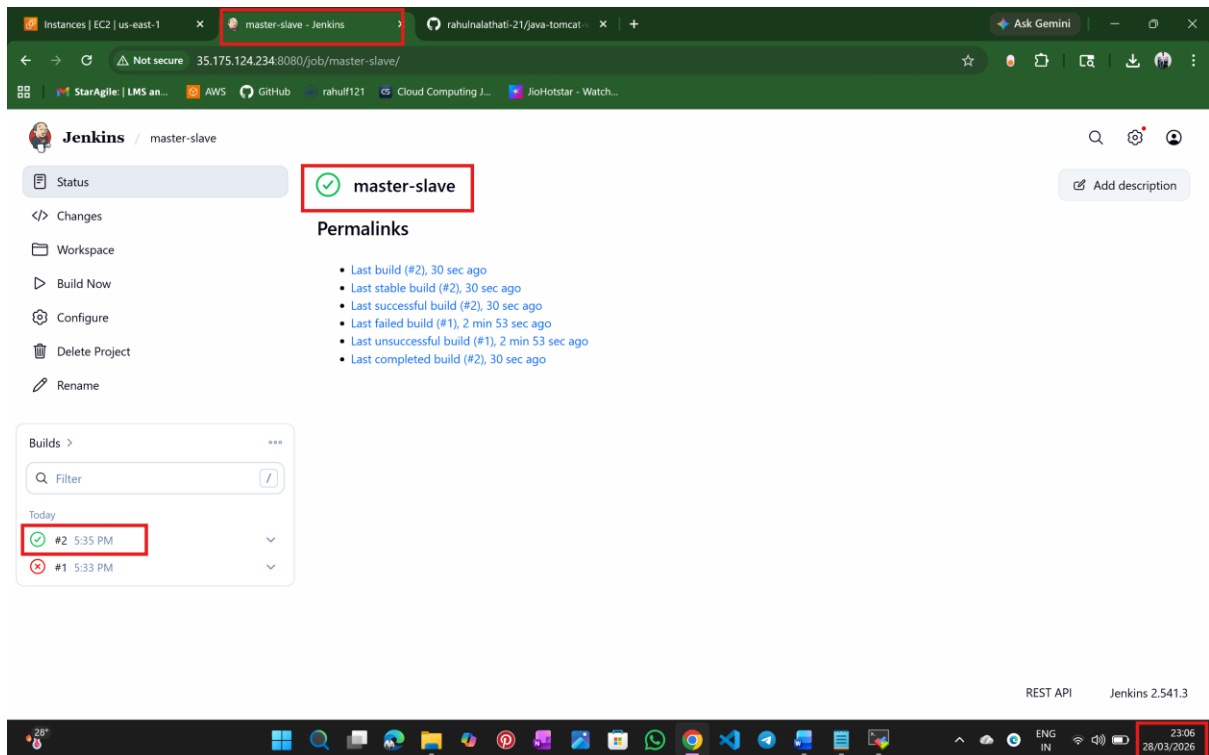
Credentials

Save Apply

23:00 28/03/2026



Step: 7 Result, so build files are store in worker node-1



Jenkins / master-slave / #2

Console

Download Copy View as plain text

Started by user **rahu**

Running as SYSTEM

Building remotely on **worker-1 (node1)** in workspace /home/ec2-user/rahu/workspace/master-slave

The recommended git tool is: NONE

No credentials specified

> git rev-parse --resolve-git-dir /home/ec2-user/rahu/workspace/master-slave/.git # timeout=10

Fetching changes from the remote Git repository

> git config remote.origin.url https://github.com/rahuinalathati-21/java-tomcat-webapp.git # timeout=10

Fetching upstream changes from https://github.com/rahuinalathati-21/java-tomcat-webapp.git

> git --version # timeout=10

> git --version # 'git version 2.50.1'

> git fetch --tags --force --progress -- https://github.com/rahuinalathati-21/java-tomcat-webapp.git +refs/heads/*:refs/remotes/origin/* # timeout=10

> git rev-parse refs/remotes/origin/main^{commit} # timeout=10

Checking out Revision dda1ce7b23c4c67d433e6a80a790f950ae6c8f19 (refs/remotes/origin/main)

> git config core.sparsecheckout # timeout=10

> git checkout -f dda1ce7b23c4c67d433e6a80a790f950ae6c8f19 # timeout=10

Commit message: "Update pom.xml"

> git rev-list --no-walk dda1ce7b23c4c67d433e6a80a790f950ae6c8f19 # timeout=10

[master-slave] \$ mvn clean package

[INFO] Scanning for projects...

[INFO]

[INFO] -----< com.devops:java-tomcat-webapp >-----

[INFO] Building Java Tomcat WebApp 1.0

[INFO] -----[war]-----

23:07 28/03/2026

worker-1

Terminal Sessions View X server Tools Games Settings Macros Help

Session Servers Tools Games Sessions View Split MultiExec Tunneling Packages Settings Help

[root@ip-172-31-11-247 ec2-user]# cd rahu/

[root@ip-172-31-11-247 rahu]#

remoting remoting.jar workspace

[root@ip-172-31-11-247 rahu]#

ip-172-31-11-247.ec2.internal 0% 0.38 GB / 0.90 GB 0.01 Mb/s 0.00 Mb/s 53 min ec2-user (x2) /: 8% /tmp: 1% /boot/efi: 13%

UNREGISTERED VERSION - Please support MobaXterm by subscribing to the professional edition here: https://mobaxterm.mobatek.net

23:11 28/03/2026

Step: 7 Result, so build files are store in worker node-1

The screenshot shows the Jenkins 'Configure' page for a job named 'master-slave'. The 'Restrict where this project can be run' checkbox is checked, and the 'Label Expression' field contains 'node2'. The 'Source Code Management' section shows 'Git' as the selected provider. The browser's taskbar at the bottom shows the time as 23:12 on 28/03/2026.

Instances | EC2 | us-east-1 | master-slave Config - Jenkins | rahulnalathati-21/java-tomcat | Ask Gemini

Not secure 35.175.124.234:8080/job/master-slave/configure

Jenkins / master-slave / Configure

Configure

- General
- Source Code Management
- Triggers
- Environment
- Build Steps
- Post-build Actions

☐ GitHub project

☐ This project is parameterized ?

☐ Throttle builds ?

☐ Execute concurrent builds if necessary ?

☒ Restrict where this project can be run ?

Label Expression ?

node2

Label node2 matches 1 node. Permissions or other restrictions provided by plugins may further reduce that list.

Advanced ▾

Source Code Management

Connect and manage your code repository to automatically pull the latest code for your builds.

☐ None

☒ Git ?

Save Apply

28°

23:12 28/03/2026

The screenshot shows the Jenkins 'master-slave' job overview page. The 'master-slave' job is highlighted with a green checkmark. The 'Permalinks' section lists various build links. The 'Builds' section shows a list of builds, with build #3 (5:43 PM) highlighted. The browser's taskbar at the bottom shows the time as 23:13 on 28/03/2026.

Instances | EC2 | us-east-1 | master-slave - Jenkins | rahulnalathati-21/java-tomcat | Ask Gemini

Not secure 35.175.124.234:8080/job/master-slave/

Jenkins / master-slave

Status

Changes

Workspace

Build Now

Configure

Delete Project

Rename

master-slave

Permalinks

- Last build (#3), 25 sec ago
- Last stable build (#3), 25 sec ago
- Last successful build (#3), 25 sec ago
- Last failed build (#1), 10 min ago
- Last unsuccessful build (#1), 10 min ago
- Last completed build (#3), 25 sec ago

Builds >

Filter

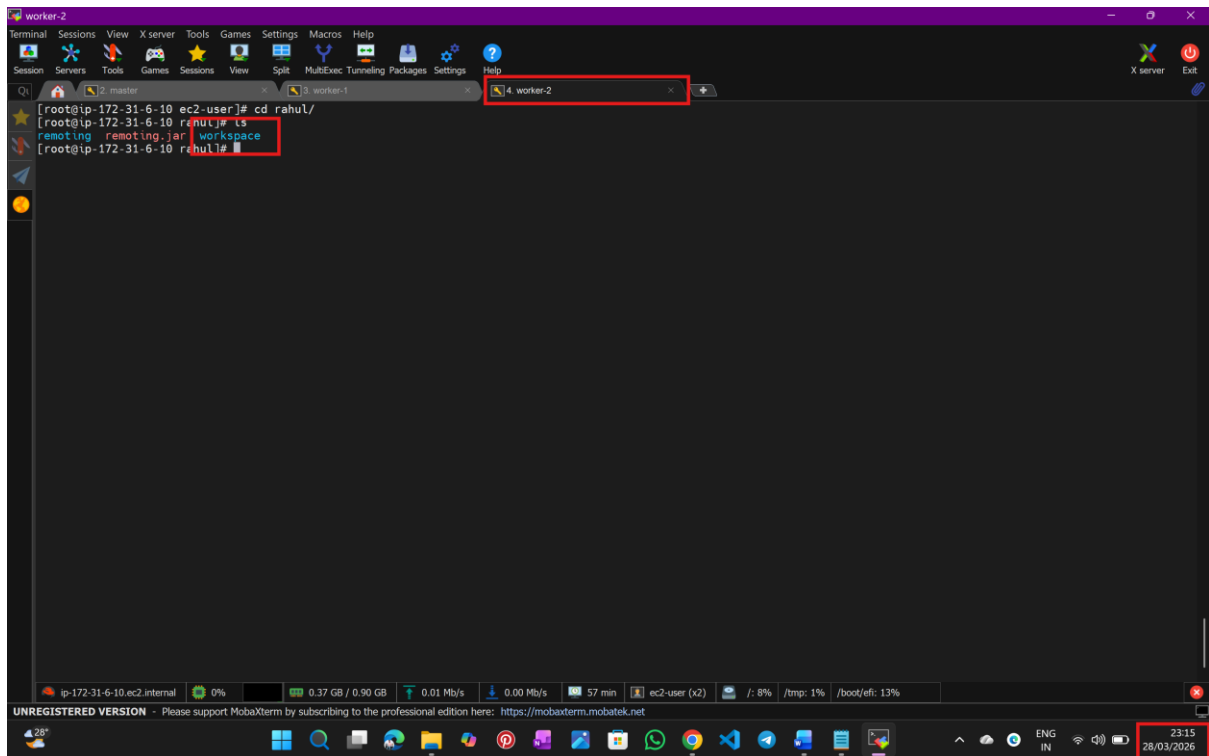
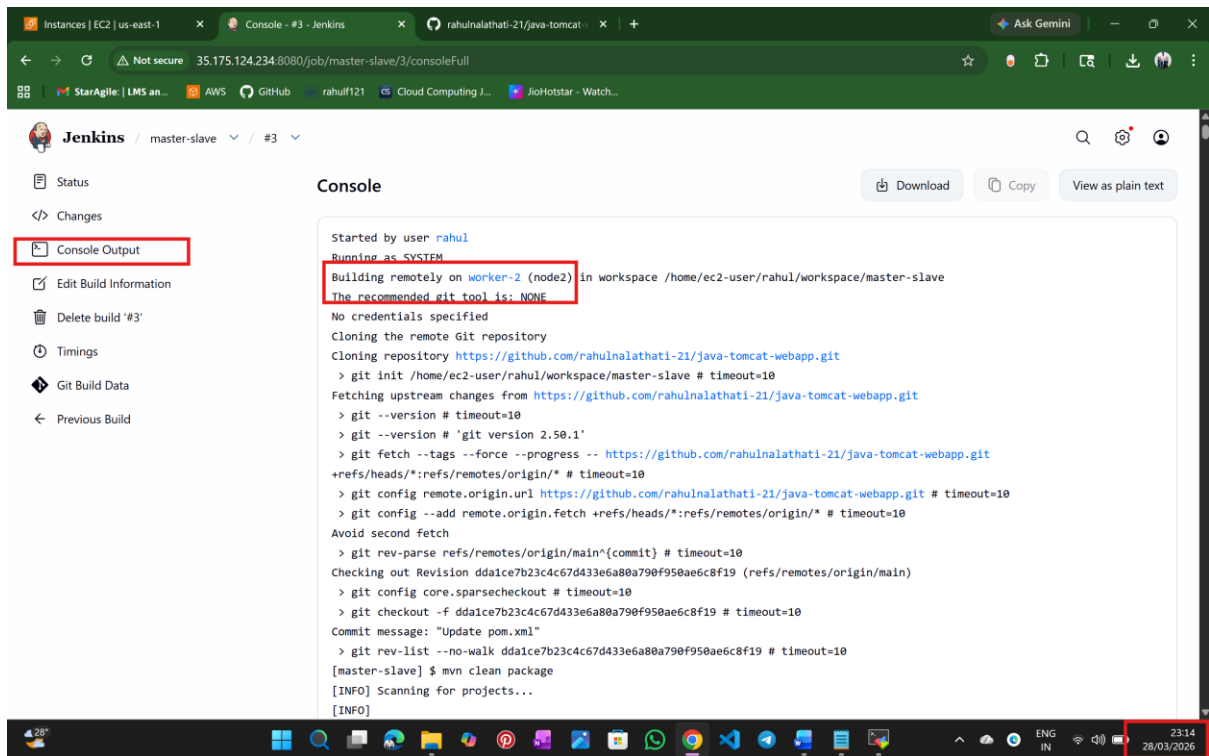
Today

- #3 5:43 PM
- #2 5:35 PM
- #1 5:33 PM

REST API Jenkins 2.541.3

28°

23:13 28/03/2026



-----END-----

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